When Does Demand Have Memory?

Empirical Characterization of AR(1) Persistence as Foundation for Adaptive Retail Allocation

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1. Motivation, Research Questions, and Data

A decade of building algorithmic recommendation systems for retail decision makers produced one consistent observation: organizations with access to better algorithms do not consistently make better decisions. The gap is not algorithmic: the models are sound. It is not a failure of intelligence: the decision makers are capable. The gap exists at the interface between the two. In commercial settings where payout structures tied revenue to experimentally verified incremental lift, this human-algorithm gap became economically measurable at scale. That observation motivates a proposed research framework I am developing: the **Economic Value of Adaptivity (EVA)**, which measures the maximum economic gain an adaptive AI-driven allocation policy delivers over a static one-shot benchmark, and the fraction of that gain realized when a human acts on the algorithm's recommendation.

In a finite-horizon MDP with T allocation epochs, three policies span the value landscape: static π_S (commits at t=0), rolling π_R (re-optimizes each epoch, capturing EVA_reactivity), and Q-function π_Q (maximizes total expected reward via the Bellman equation, capturing EVA_reactivity + EVA_anticipation). EVA decomposes as:

$$\mathrm{EVA}_{\mathrm{reactivity}} = \mathbb{E}\left[\Pi_{\pi_R} - \Pi_{\pi_S}\right] \quad \mathrm{EVA}_{\mathrm{anticipation}} = \mathbb{E}\left[\Pi_{\pi_Q} - \Pi_{\pi_R}\right] \text{ and } \mathrm{EVA}_{\mathrm{total}} = \mathbb{E}\left[\Pi_{\pi_Q} - \Pi_{\pi_S}\right]$$

The EVA framework assumes demand D_t follows a **log-Normal process**, strictly positive and right-skewed, consistent with retail sales data where moderate weekly volumes are punctuated by occasional demand surges. Formally: $\log(D_t) = \mu + \rho \cdot \log(D_{t-1}) + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}\left(0, \sigma_\eta^2\right)$

It is hypothesized that the relative magnitude of EVA_reactivity versus EVA_anticipation depends on ρ, the AR(1) persistence parameter measuring how much this week's demand predicts next week's, though the formal derivation is reserved for the simulation study. AR(1) is selected over higher-order models because sample autocorrelation functions (ACFs) — which measure the correlation between a time series and its own lagged values at successive time steps — show significant lag-1 correlation with rapid decay beyond lag-1, consistent with a first-order process.

Research Questions

RQ1: Is AR(1) demand persistence statistically significant and economically meaningful across the continuous retail SKU catalog, or is weekly demand effectively memoryless? **RQ2:** Is the persistence parameter ρ a SKU-level property, or is it predictable from product category taxonomy? **RQ3:** Does ρ remain stable over a multi-year horizon, or does it exhibit regime shifts consistent with the burst-and-decay and progressive amplification demand environments posited in the EVA typology?

Data and Demand Environment Typology

The **M5 Walmart Hierarchical Sales dataset** was selected to answer these questions because it provides the richest publicly available longitudinal retail scanner data at SKU level: 5.3 years of daily unit sales across 30,490 SKU-store combinations with documented price, promotion, and calendar event metadata, enabling rigorous assumption testing without proprietary data access. Its hierarchical structure supports both SKU-level estimation (RQ1 and RQ2) and cross-store validation (RQ3) within a single framework.

The EVA simulation uses four demand environments, theoretically motivated by the persistence and shock structure of demand, with the following SKUs serving as empirical anchors. These are illustrative characterizations; the behavioral mechanisms driving low or high ρ are likely multi-causal and the patterns below represent plausible explanations rather than definitive accounts:

DE1: Low persistence (ρ < 0.3). e.g., HOUSEHOLD_2_297 (inferred: specialty personal care, $\hat{\rho} = 0.114$). Low measured ρ may reflect repurchase cycles longer than the weekly window, substitution behavior across brands, heterogeneous purchase timing across a large customer base averaging to near-zero autocorrelation, or price-driven

opportunistic purchasing. Demand shocks tend to be supply-driven and self-correcting. EVA_reactivity is likely to dominate.

DE2: Moderate persistence ($0.3 \leq \rho < 0.6$). e.g., FOODS_2_305 (inferred: frozen meals or yogurt, $\hat{\rho} = 0.493$). Weekly grocery trip habit drives moderate persistence; customers who incorporate this SKU into their basket tend to continue. Trial-inducing promotions create temporary ρ elevation that decays as lapsed customers exit. The rolling policy captures most available value.

DE3: Intermittent burst with ρ collapse. e.g., HOBBIES_1_342 (inferred: seasonal fitness accessory, ρ shifts 0.780 → −0.013). Occasion-driven purchasing: a discrete external trigger activates a customer cohort simultaneously, creates a sharp demand spike, then dissipates. No carry-forward habit forms. State detection, specifically knowing whether the SKU is in burst or dormant phase, is the primary value driver.

DE4: Structural break with ρ amplification. e.g., HOBBIES_1_023 (inferred: outdoor/exercise item, ρ shifts 0.473 → 0.880). Progressive habit formation through sequential adoption waves: a new product, reformulation, or competitor exit introduces the SKU to successive customer cohorts who habituate incrementally, creating a staircase demand pattern. *Note: M5 anonymizes product identities; inferences based on Walmart's published department taxonomy.*

2. Data Quality and Exploratory Analysis

The analysis focuses on store CA_1 (3,049 SKUs, 274 weeks) with cross-store validation across all 10 stores. Five deliberate data quality choices were made (Table 1, Appendix): daily sales were aggregated to weekly to remove day-of-week seasonality while preserving the natural retail planning cycle; a 70% non-zero threshold was applied because $\log(D_t)$ is undefined at zero; sales were $\log(1+D_t)$ transformed to stabilize right-skew; estimation was anchored to CA_1 to keep modeling assumptions auditable; and a stratified random sample of 15 SKUs per category (45 total) was drawn for analytical depth.

[Table 1 (Appendix): *Data Quality Decisions*]

EDA confirmed three specification choices (Table 2, Appendix): raw demand is right-skewed and log1p transformation produces approximate symmetry, supporting the log-Normal assumption; 91% of SKUs show significant lag-1 autocorrelation with rapid decay beyond lag-1, justifying AR(1) over higher-order models; and three observations exceeding 5σ in log-space all coincide with documented promotional price drops exceeding 30% and were retained as genuine demand responses. Of 3,049 CA_1 SKUs, 1,639 (53.8%) fail the 70% non-zero filter but account for only 24.3% of revenue; the continuous AR(1) model covers 75.7% of store revenue. Intermittent SKUs, concentrated in FOODS_3 (deli and prepared foods), require a separate model class outside this study's scope.

[Table 2 (Appendix): *EDA Summary*]

3. Statistical Analysis

3.1.1. RQ1: AR(1) Estimation using OLS Regression and Hypothesis Testing

Methodology. The AR(1) model $\log(1+D_t) = \mu + \rho \cdot \log(1+D_{t-1}) + \varepsilon_t$ was estimated by OLS (n = 273 pairs per SKU). H_0: $\rho = 0$ was tested per SKU via t-test. SKUs were classified into Low ($\rho < 0.3$), Medium ($0.3 \leq \rho < 0.6$), and High ($\rho \geq 0.6$) tiers, validated by testing H_0: $\rho = 0.5$ per tier: well-chosen boundaries should place the High mean clearly above 0.5, the Low clearly below, and the Medium not significantly different from 0.5 (Tables 3 and 4, Appendix).

[Table 3 (Appendix): *AR(1) Estimates by Category*]

[Table 4 (Appendix): *Persistence Tier Validation*]

Findings. 41 of 45 SKUs (91.1%) reject H_0: $\rho = 0$ at $p < 0.05$ (mean $\hat{\rho} = 0.550$, 95% CI [0.457, 0.643]). Tier distribution: High = 23, Medium = 13, Low = 9. The High tier rejects H_0: $\rho = 0.5$ at $t = 12.691$ ($p < 0.001$, mean $R^2 = 0.574$); the Low tier at $t = -8.924$ ($p < 0.001$, mean $R^2 = 0.030$); the Medium tier fails to reject ($p = 0.534$), confirming correct centering within the 0.3–0.6 boundary range. **RQ1 is answered affirmatively: demand persistence is real, widespread, and carries substantial explanatory power.** Behaviorally, high ρ reflects the repeat purchase tendency of customers; habitual buying creates autocorrelation, meaning a strong week signals the same customers are likely to return. Low ρ captures two distinct patterns: opportunistic or event-driven purchases

where the trigger is unlikely to repeat, and habitual purchases with repurchase cycles longer than the weekly observation window, producing low measured ρ not because behavior is irregular but because the measurement frequency is mismatched to the purchase rhythm.

[Figure 1: *SKU-level ρ ranked with 95% CIs and tier coloring (Appendix)*]

3.1.2. RQ1: Validating the Log-Normal AR(1) Assumption using Shapiro-Wilk and Kolmogorov-Smirnov Tests

Methodology. AR(1) residuals tested for normality using two complementary tests. SW is the most powerful test for normality departures in small-to-medium samples but becomes over-sensitive at large n, detecting negligible deviations as significant. KS tests whether the overall shape of standardized residuals matches N(0,1), which is less powerful but more robust to the sample-size inflation that makes SW unreliable at $n \approx 273$. Running both provides a high-sensitivity read (SW) and a conservative, practically relevant read (KS); when they disagree, a power simulation resolves which to trust (Table 5, Appendix).

[Table 5 (Appendix): *Goodness-of-Fit Results by Category*]

Findings. SW rejects normality for all 45 SKUs. A power simulation establishes 46.5% false rejection for near-normal t(10) draws at n = 273 versus 16.7% at n = 50, confirming this is a power artifact. The KS test passes for 24.4% of SKUs. Q-Q plots (Figure 3) show tight central-mass fit with moderate left-tail departure. The Gaussian assumption systematically underrepresents the frequency of high-impact demand events such as promotional surges, competitor stockouts, and holiday spikes that generate residuals in the tails more often than $N(0, \sigma^2)$ predicts. **The log-Normal AR(1) is a defensible working specification for the central mass of demand; tail behavior requires the extended error structure in Section 4.**

[Figure 3: *Q-Q plots: AR(1) residuals for one representative SKU per tier (Appendix)*]

3.2. RQ2: One-Way ANOVA with Levene's Test for Homogeneity of Variances

Methodology. H_0: μ_FOODS = μ_HOBBIES = μ_HOUSEHOLD tested by one-way ANOVA. ANOVA's F-test assumes equal variance across groups; if one group's ρ values are much more spread than another's, the F-statistic is unreliable. Levene's test checks this assumption before the result can be trusted. Levene yields L = 0.116 (p = 0.891), confirming homogeneous variances. Effect size: η^2 = SS_between / SS_total.

Findings. F = 0.979, p = 0.384, $\eta^2 = 0.044$. Category explains 4.4% of variance in ρ. Within-category SDs (0.249–0.270) dwarf between-category differences; a FOODS SKU can have ρ = 0.05 or ρ = 0.88, making the category label statistically uninformative. **RQ2 is answered: ρ is a SKU-level property.** Behaviorally, demand persistence reflects customer-level attachment to specific SKUs, not product taxonomy. A consumer who habitually buys one cleaning product may purchase a different HOUSEHOLD item only opportunistically. This null result is the methodological pivot: demand environment classification must be estimated at the SKU level.

[Figure 2: *ANOVA boxplot with CI overlay and overlaid ρ histograms by category (Appendix)*]

3.3.1. RQ3a: Chow Test for Structural Breaks: Temporal Stability of ρ

Background. Temporal stability of ρ means the persistence parameter does not shift across sub-periods of the observation window. If ρ is temporally stable, the AR(1) model fitted on 5.3 years reliably describes demand at any point in that window. If not, the full-series estimate is a time-averaged blend of two genuinely different regimes, like fitting one regression line through data with two distinct slopes, producing a ρ estimate that accurately describes neither sub-period and calibrates any policy built on it to a demand environment that never actually existed.

Methodology. H_0: the AR(1) slope and intercept are equal in the first and second halves of the series, meaning a single model describes the full horizon. Rejection implies a structural regime change. Tested for all 23 high-ρ SKUs via Chow test at midpoint (week 137): $F = [(RSS_full - RSS_1 - RSS_2)/k] / [(RSS_1 + RSS_2)/(n-2k)]$, k=2, $F \sim F(k, n-2k)$ under H_0. The Chow test is an F-test on restricted versus unrestricted regression, following the same logic as testing whether a regression slope differs across two subgroups (Table 6, Appendix).

[Table 6 (Appendix): *Structural Break Summary: Chow Test on High-ρ SKUs*]

Findings. 18 of 23 high-ρ SKUs (78.3%) reject H_0, meaning most high-persistence SKUs do not maintain a temporally stable demand structure over the full horizon. This is counterintuitive but consequential: high measured ρ over five years does not mean persistently high ρ throughout. The full-series estimate masks two genuinely different

sub-period regimes, and a policy calibrated on it is optimised for a demand environment that no longer exists in the second half of the horizon. Two patterns emerge: ρ collapse (14 SKUs) reflects decay of customer habit formation, specifically trend death, price disruption, or substitute entry; ρ amplification (4 SKUs) reflects progressive adoption, specifically successive customer cohorts habituating through promotion, endorsement, or competitor disruption. **RQ3 is resolved: temporal stability is absent for 78.3% of high-ρ SKUs, and the observed shift patterns map directly to the DE3 and DE4 regime structures posited in the EVA typology.**

[Figure 4: *ρ pre- vs. post-break for 18 Chow-detected break SKUs (Appendix)*]

3.3.2. RQ3b: Two-Way ANOVA with RCBD: Cross-Store Stability of ρ

Methodology. AR(1) was estimated for 45 sampled SKUs across all 10 stores (349 of 450 pairs estimable; remainder intermittent). The generalization question is a blocking problem: SKU-level heterogeneity in ρ will mask any store effect unless controlled for. The RCBD design treats SKUs as blocks and stores as treatments, isolating the store effect after removing SKU-level variation, following the same logic as blocking on a nuisance variable to increase sensitivity to the treatment of interest. Model: $\rho_{ij} = \mu + \alpha_i$ (SKU block) + β_j (Store treatment) + ε_{ij}. A one-way ANOVA on store alone, without blocking on SKU, was also computed as a counterfactual: if the RCBD design adds value, the store effect should be more visible after blocking than before it (Table 7, Appendix).

[Table 7 (Appendix): *Two-Way ANOVA: ρ ~ SKU + Store (RCBD)*]

Findings. SKU explains 70.5% of ρ variance ($F = 17.896$, $p < 0.001$, $\eta^2 = 0.705$); Store explains 3.1% ($F = 3.802$, $p < 0.001$, $\eta^2 = 0.031$). The value of blocking is directly demonstrated: one-way ANOVA on store alone fails to reject H_0 ($F = 1.191$, $p = 0.300$) because SKU heterogeneity masks the store signal; blocking raises the store F from 1.19 to 3.80. Within-SKU ρ range across stores has median 0.326; 53.3% of SKUs exceed the 0.3 tier-crossing threshold. The store effect is best understood as a customer mix effect: TX_1 (mean ρ = 0.414) versus TX_3 (0.562) serve the same SKU catalog but different customer populations; TX_1's catchment appears more opportunistic and less habitual, possibly due to demographics, competitor proximity, or a higher transient customer share.

[Figure 5: *ρ by store with CI and within-SKU cross-store range (Appendix)*]

4. Updated Demand Specification and Validation

Original simulation prior: $\log(D_t) = \mu + \rho \cdot \log(D_{t-1}) + \varepsilon_t$, $\varepsilon_t \sim N(0, \sigma^2)$, $\rho \in [0,1)$ fixed per SKU. The empirical analysis motivates two refinements, yielding the updated demand generating process:

$$\log(D_{t,s}) = \mu(s) + \rho(s,t) \cdot \log(D_{t-1,s}) + \varepsilon_{t,s}$$
$$\rho(s,t) = \rho_0(s) + \gamma \cdot \delta_{\text{store}} + \mathbf{1}(t > \tau) \cdot \Delta\rho(s)$$
$$\varepsilon_{t,s} \sim t\big(\nu(s),\ 0,\ \sigma^2(s)\big)$$

where $\boldsymbol{\rho_0(s)}$ is the baseline SKU-level persistence estimated by OLS (empirical distribution: mean 0.550, SD 0.260); $\boldsymbol{\gamma \cdot \delta_store}$ is the store-level adjustment capturing the 3.1% of ρ variance attributable to customer mix differences; $\boldsymbol{\mathbf{1}(t > \tau) \cdot \Delta\rho(s)}$ is the regime shift term where τ is the structural break point, Δρ is negative for DE3 collapse SKUs (mean −0.531) and positive for DE4 amplification SKUs (mean +0.365); and $\boldsymbol{\nu(s)}$ is the SKU-level degrees of freedom governing tail weight, converging to Gaussian as $\nu \to \infty$. Table 9 (Appendix) summarizes the full before-and-after comparison.

Refinement 1: Time-varying, store-adjusted persistence. The store adjustment $\gamma \cdot \delta_store$ is not a statistical nuisance; it signals that different stores serve customer populations with distinct purchasing rhythms. A discrete choice model capturing repurchase probability as a function of prior purchase history, price sensitivity, and local demographics would provide the structural explanation for why $\gamma \cdot \delta_store$ varies across stores.

Refinement 2: Heavier-tailed errors. The t distribution is symmetric and bell-shaped like the Gaussian but with heavier tails governed by ν; lower ν means fatter tails and more frequent extreme values, converging to $N(0, \sigma^2)$ as $\nu \to \infty$. Estimating ν per SKU from residual tail behavior allows the model to adapt: a SKU prone to promotional spikes or competitor disruptions receives a low ν, honestly reflecting how often extreme demand events occur.

Specification validation. RSS reduction, not R^2, is the appropriate comparison metric when splitting a series, because period-level R^2 computes TSS against its own within-period mean rather than the full-series mean, making it non-

comparable across specifications. The two-period model reduces RSS by 10.0% for break-detected SKUs versus 3.2% for non-break SKUs; the 6.8 percentage point difference is attributable to genuine regime change, not overfitting. The updated specification is a materially better representation of demand for 78.3% of high-ρ SKUs. Its current limitation is that it requires knowing the break point τ in advance. The natural next step is break detection from the data itself, a method that monitors the demand series and identifies when evidence for regime change crosses a statistical threshold, enabling the model to adapt as structural shifts occur rather than retrospectively. Full results in Table 8 (Appendix).

[Table 8 (Appendix): *Model Fit: Static vs. Updated Specification*]
[Table 9 (Appendix): *Updated Demand Specification: Before and After Comparison*]

5. Implications

SKU-level estimation is necessary. Category explains 4.4% of ρ variance. Taxonomy-based demand environment assignment would misclassify most of the assortment. The EVA framework must estimate ρ from observed demand trajectories at the SKU level.

The DE3 and DE4 regime structures are empirically present. The four ρ-amplification SKUs confirm that progressive habit formation exists in real retail data; DE4 is not a theoretical construct. The 14 collapse SKUs confirm DE3's burst-and-decay structure. Both are valid inputs to the EVA simulation design.

The empirical ρ distribution should inform the simulation parameter grid. Real SKUs concentrate in the high-persistence range (mean $\rho = 0.550$, 51.1% above $\rho = 0.6$) and post-break amplification reaches $\rho = 0.884$. The EVA simulation grid should be weighted toward this region rather than spread uniformly across the theoretical [0,1) range.

Store-level recalibration is required for deployment. 53.3% of SKUs show cross-store ρ range exceeding 0.3, sufficient to shift tier classification. A single global ρ estimate per SKU cannot support production deployment across multiple store formats.

Generalization boundary. Results are drawn from 45 SKUs in one Walmart store format. RCBD validation spans 10 stores in 3 states but findings should not be extrapolated to other retail banners or geographies without additional calibration.

6. Conclusion

What was found. AR(1) persistence is statistically significant in 91.1% of sampled SKUs (mean $\rho = 0.550$), answering RQ1 affirmatively. One-way ANOVA finds no category effect ($F = 0.979$, $p = 0.384$, $\eta^2 = 0.044$), answering RQ2: ρ is SKU-specific. Chow tests reveal 78.3% of high-ρ SKUs exhibit structural regime change, with ρ collapsing in 60.9% and amplifying in 17.4%, resolving RQ3: temporal stability is the exception, not the norm. Cross-store RCBD confirms ρ is primarily a SKU property ($\eta^2 = 0.705$) with a significant store secondary effect ($\eta^2 = 0.031$) invisible without blocking.

What was inferred. Demand persistence reflects customer habit formation at the SKU level. Its non-stationarity maps to identifiable real behaviors: habit decay (ρ collapse) and progressive customer adoption (ρ amplification). The Gaussian error assumption understates tail event frequency, making EVA_anticipation estimates conservative lower bounds. Store-level ρ differences reflect customer mix, not product characteristics alone.

Next steps. Three directions follow. First, extend from 45 sampled SKUs to the full 1,410 eligible CA_1 SKUs and replicate across store formats to establish broader generalizability. Second, develop break detection from the data itself rather than at a fixed midpoint, a prerequisite for the EVA simulation to model DE3 and DE4 environments realistically. Third, develop a discrete choice model of customer repurchase behavior to provide structural grounding for why ρ varies across SKUs and stores, moving from statistical description toward a behavioral explanation that can inform both the EVA simulation design and its eventual field deployment.

Note on LLM Usage.

Developed with the assistance of Claude (Anthropic, claude-sonnet-4-6) for framework structuring, Python code validation and tuning across nine analysis blocks, and iterative drafting of the report. All analytical decisions, test choices, interpretations, and framework connections were made by the author. Code, data pipeline, and annotated notebooks: github.com/pinakighosh/sta611-demand-persistence

Appendix: Figures and Tables

Figure 1. *SKU-Level ρ Estimates Ranked by Persistence Tier.*

Blue = Low, orange = Medium, green = High. Continuous gradient confirms persistence is a spectrum; CI separation validates tier boundaries.

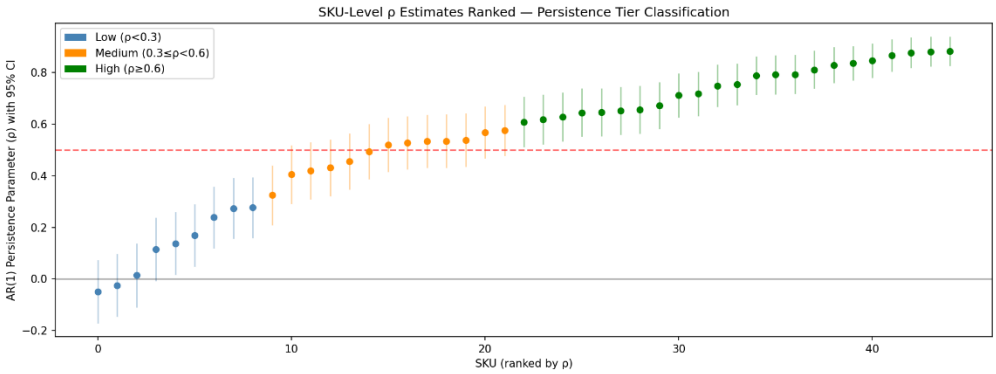

Figure 2. *One-Way ANOVA: ρ by Product Category.*

Boxplot with group mean and 95% CI (left); overlaid histograms (right). Near-complete overlap confirms F = 0.979, p = 0.384.

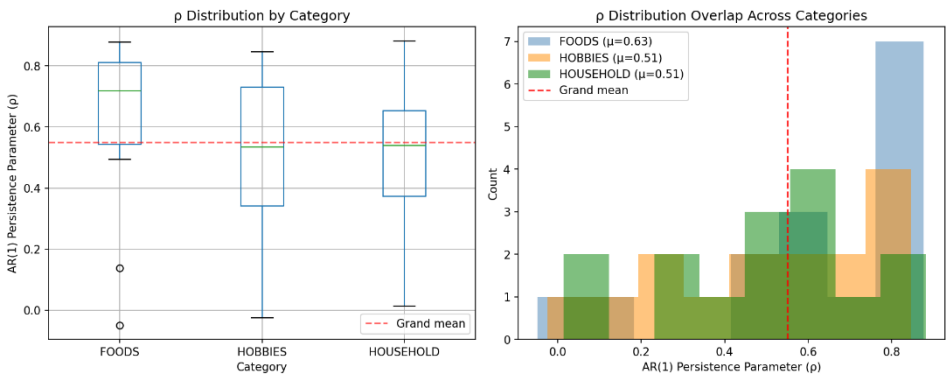

Figure 3. *Q-Q Plots: AR(1) Residuals for Representative SKUs.*

One SKU per persistence tier. Tight central-mass fit with moderate left-tail departure supports log-Normal AR(1) as a defensible working specification.

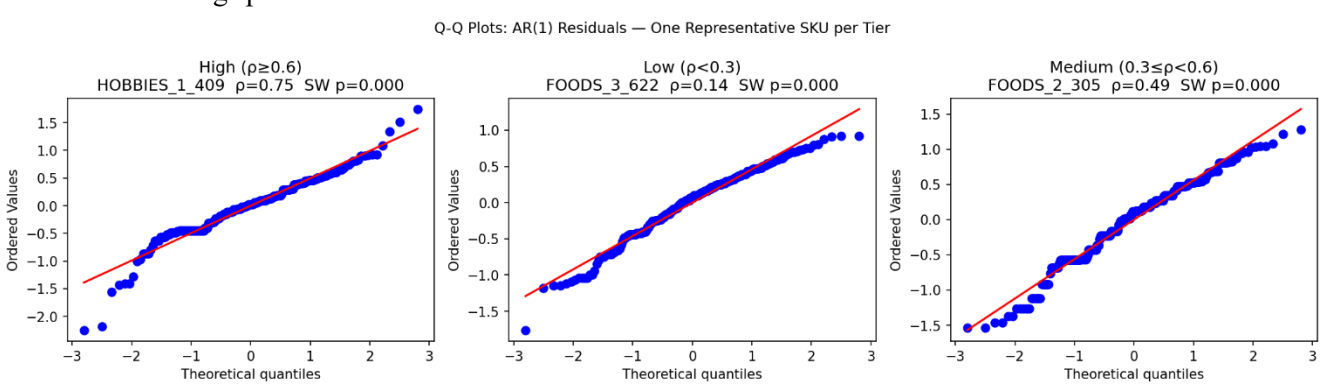

Figure 4. *Structural Break: ρ Pre- vs. Post-Break for 18 High-ρ SKUs.*

Blue = pre-break $\hat{\rho}$; red = post-break. Dominant downward pattern = ρ collapse (DE3); four upward cases = ρ amplification (DE4).

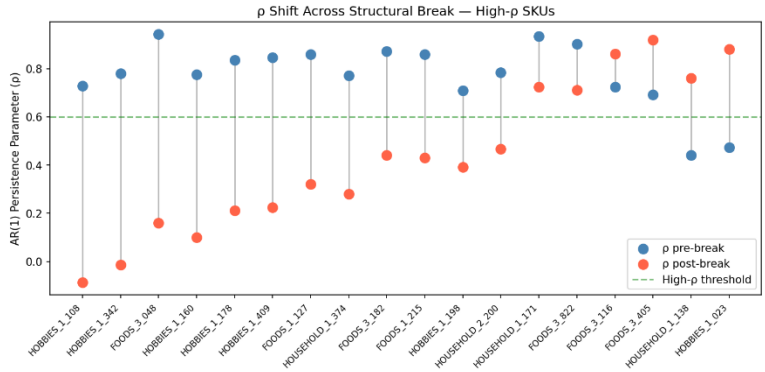

Figure 5. *Cross-Store Generalization: Two-Way ANOVA (RCBD).*

Left: ρ by store with mean and 95% CI. Right: Within-SKU ρ range across 10 stores. Blocking on SKU reveals a store effect invisible to one-way ANOVA.

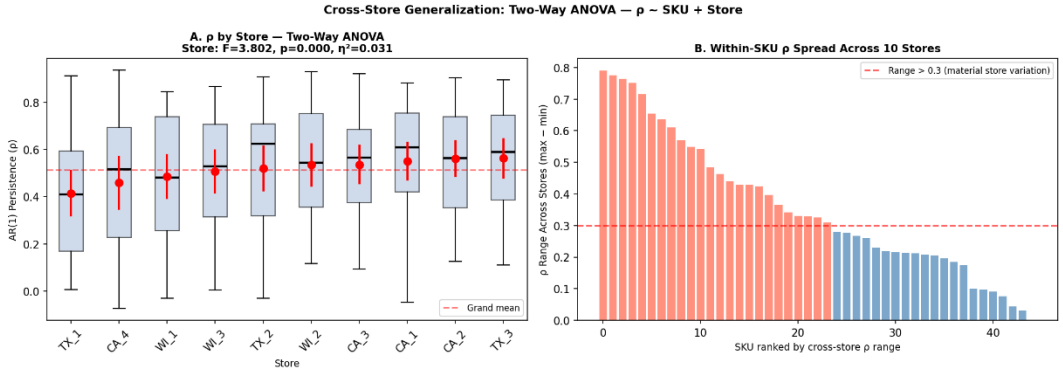

Table 1. *Data Quality Decisions: Choices, Justifications, and Risks.*

Decision	Choice	Statistical Justification	Alternative & Risk
Temporal unit	Weekly aggregation (274 wks)	Day-of-week effects inflate lag-1 ACF; week = natural allocation epoch and the industry-standard retail replenishment planning cycle; buyers review and adjust weekly, not daily	Daily: spurious AR(1) from weekday seasonality
Zero-demand filter	70% non-zero; retains 1,410 of 3,049 SKUs	$\log(D_t)$ undefined at $D_t=0$; AR(1) requires continuous demand	Stricter → loses SKUs; looser → intermittency biases $\hat{\rho}$
Transformation	$\log(1+D_t)$; log1p handles residual zeros	Stabilises right-skew; ρ retains autocorrelation interpretation	Raw sales: heteroscedastic residuals inflate SE($\hat{\rho}$)
Store scope	CA_1 primary; all 10 for RCBD validation	Single-store keeps Gauss–Markov conditions auditable	Pooling requires hierarchical model, addressed partially by RCBD
Sample	Stratified random, 15 SKUs/category (seed=42)	Equal category representation; reproducible; no cherry-picking	Census (1,410 SKUs): limits per-SKU analytical depth

Table 2. *EDA Summary: Checks Conducted Prior to Model Fitting.*

EDA Check	Finding	Method	Implication
Distributional shape	Raw demand right-skewed; log1p ≈ symmetric	Histogram + Q-Q pre-transform	log-Normal assumption defensible post-transformation
Serial correlation	Significant lag-1 ACF in 91% of SKUs; rapid decay beyond lag-1	Sample ACF per SKU (274 weeks)	AR(1) preferred over AR(p>1)

EDA Check	Finding	Method	Implication
Outlier detection	3 obs > 5σ in log-space; all coincide with price drops > 30%	5σ rule; cross-ref sell_prices.csv	Retain: genuine demand response, not recording error
Event effects	Holiday weeks show elevated FOODS demand	M5 calendar event flags	No exclusion; tail events are cases EVA_anticipation serves
Intermittent scope	1,639 SKUs (53.8%) fail filter; only 24.3% of revenue	Non-zero fraction; revenue-weighted share	Continuous model covers 75.7% of revenue

Table 3. *AR(1) Persistence Estimates by Category.*

Category	n	Mean $\hat{\rho}$	SD	95% CI	% reject H_0:ρ=0
FOODS	15	0.626	0.270	[0.477, 0.776]	100.0%
HOBBIES	15	0.509	0.260	[0.365, 0.653]	86.7%
HOUSEHOLD	15	0.514	0.249	[0.376, 0.652]	86.7%
All SKUs	45	**0.550**	0.260	[0.457, 0.643]	**91.1%**

Table 4. *Persistence Tier Validation: t-tests against H_0: ρ = 0.5.*

Persistence Tier	n	t vs H_0:ρ=0.5	p-value	Mean R^2
High (ρ ≥ 0.6)	23	**12.691**	**< 0.001**	**0.574**
Medium (0.3 ≤ ρ < 0.6)	13	−0.641	0.534	0.242
Low (ρ < 0.3)	9	**−8.924**	**< 0.001**	**0.030**

Table 5. *Goodness-of-Fit Results by Category.*

Category	n	SW pass rate	KS pass rate	Tail pattern	Assessment
FOODS	15	0/15 (0%)	3/15 (20%)	Mild left-tail	Defensible working approximation
HOBBIES	15	0/15 (0%)	6/15 (40%)	Closest to Gaussian	Defensible
HOUSEHOLD	15	0/15 (0%)	2/15 (13%)	Moderate left-tail	Defensible working approximation
All SKUs	45	0/45 (0%)	**11/45 (24%)**	Log-left tail	Working approximation
Power sim: t(10) at n=273	N/A	46.5% false reject	16.7% at n=50	Same distribution	SW over-rejects at n≈273

Table 6. *Structural Break Summary: Chow Test on High-ρ SKUs.*

Group	n tested	Break detected	Mean ρ pre	Mean ρ post	Mean Δρ	Illustrative SKU (inferred type)
All high-ρ SKUs	23	18 (78.3%)	0.800	0.457	−0.342	N/A
ρ collapse (DE3 analog)	23	14 (60.9%)	0.821	0.290	−0.531	HOBBIES_1_342: seasonal fitness; 0.780→−0.013
ρ amplification (DE4 analog)	23	4 (17.4%)	0.519	0.884	+0.365	HOBBIES_1_023: outdoor/exercise; 0.473→0.880
No break detected	23	5 (21.7%)	0.743	0.711	−0.032	HOUSEHOLD_1_171: stable ρ≈0.88

Table 7. *Two-Way ANOVA: ρ ~ SKU + Store (RCBD, SKU as Block).*

Source	SS	df	MS	F	p-value	η^2
SKU Identity	16.037	44	0.365	**17.896**	**< 0.001**	**0.705**

Source	SS	df	MS	F	p-value	η^2
Store Identity	0.697	9	0.077	3.802	< 0.001	0.031
Residual	6.008	295	0.020	N/A	N/A	0.264
Total	22.742	348	N/A	N/A	N/A	**1.000**

Table 8. *Model Fit Comparison: Static vs. Updated Specification.*

Specification	n SKUs	Mean R^2	RSS Reduction	Incremental vs Baseline	Interpretation
Static AR(1), full series, all 45 SKUs	45	0.369	N/A	N/A	Single ρ per SKU, full horizon (simulation prior)
Static AR(1), high-ρ tier only	23	0.574	N/A	N/A	Persistence signal concentrated in high-ρ tier
Two-period AR(1), non-break SKUs (noise floor)	27	0.196	3.2%	N/A	Expected noise from arbitrary midpoint split
Two-period AR(1), break-detected SKUs	18	0.448	**10.0%**	**+6.8 pp attributable to regime change**	Updated spec reduces unexplained variance
RCBD: SKU + Store, 349 pairs	349	N/A	N/A	$\eta^2 = 0.705 + 0.031 =$ 73.6% variance explained	Store recalibration adds incremental validity

Table 9. *Updated Demand Specification: Before and After Comparison.*

Component	Original Simulation Prior	Updated Specification	Empirical Basis
ρ structure	Fixed scalar ρ per SKU, constant over horizon	$\rho(s,t) = \rho_0(s) + \gamma \cdot \delta_store + 1(t>\tau) \cdot \Delta\rho(s)$	78.3% of high-ρ SKUs show structural breaks; 3.1% store variance
Error distribution	$N(0, \sigma^2)$ — symmetric Gaussian	$t(\nu, 0, \sigma^2)$, ν estimated per SKU from residual tail behavior	GOF: moderate left-tail departures across all categories
ρ parameter space	Uniform over [0,1)	Empirically calibrated: mean 0.550, 51.1% of continuous SKUs above ρ = 0.6	Full AR(1) estimation across 45 SKUs, 274 weeks
Regime structure	None — single stationary process	Break point τ with Δρ: negative for DE3 (mean −0.531), positive for DE4 (mean +0.365)	Chow test: 18 of 23 high-ρ SKUs reject temporal stability
Store adjustment	Not modeled	γ·δ_store: store-level shift in ρ reflecting local customer mix	Two-way ANOVA RCBD: $\eta^2_store = 0.031$, $F = 3.802$, $p < 0.001$